

Pranav Shridhar Kowadkar

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Data Scientist and Software Engineer with 3+ years of experience creating scalable ML solutions, refining data pipelines, and optimizing cloud-based infrastructures. Skilled in Python, TensorFlow, AWS, Spark, and Airflow, consistently delivering measurable efficiency gains and improved system performance.

TECHNICAL SKILLS

- Programming & Visualization: Python, R, C/C++, C#, Java, JavaScript, SQL, NoSQL, SAS, SPSS, Tableau, PowerBI, Looker Studio
- Frameworks & Tools: TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, .NET, Spark, PySpark, Spark ML, Airflow
- Data Science & ML: Machine Learning, Deep Learning, Data Analysis, Predictive Modeling, Data Mining, Prompt Engineering, NLP
- Cloud Technologies: AWS (S3, RDS, Lambda, CloudFront, SageMaker, EMR, Kinesis), Azure, Databricks
- Big Data & Distributed Systems: Apache Spark, Hadoop, Kafka, Distributed Computing, Data Processing at Scale
- Data Engineering: Elastic search, Redis, ETL pipelines, Data Warehousing, Data Lakes, Delta Lake

PROFESSIONAL EXPERIENCE

Programmer Analyst | Vaandu, Newark, NJ | Python, Oracle, Unix, SQL, AWS Nov 2024 - Present

- Strengthened **Oracle-driven data pipelines** using **Unix scripting**, boosting query speeds by 20% and ensuring smoother batch processes
- Enhanced **Python** code quality through targeted refactoring, cutting review times by 15% and improving deployment stability

Data Scientist | JerseySTEM, Florham Park, NJ | Python, SQL, GitHub (CI/CD), Airflow Feb 2024 - Nov 2024

- Implemented advanced **SQL queries** that streamlined data manipulation processes, allowing for efficient analysis of **1M+ data points**
- Automated reporting processes with **Airflow**, cutting manual efforts by 40% and improving efficiency by 25% across operational tasks
- Championed Airflow adoption by training 10+ team members and **integrating workflows**, achieving a 40% boost in **task automation**

Data Engineer | Bayer, Remote, NJ | Python, SQL, Data Analytics, Azure, Databricks, Snowflake Sep 2023 - Dec 2023

- Architected Databricks pipelines to process Snowflake data, generating 3 personas for targeted marketing, ensuring data accuracy, reliability
- Developed **PowerBI dashboards** to visualize 121+ behavior metrics, improving campaign efficiency with data-driven insights
- Enhanced **ETL processes** for real-time **POS data**, using the 3 key metrics to optimize advertising and boost campaign performance

Teaching Assistant | New Jersey Institute of Technology, NJ | Big Data, Data Structures, Physics Feb 2023 - Dec 2023

- Mentored 150+ students in **General Physics**, **Big Data**, and **Data Structures** courses, providing instruction and evaluating student work

R&D Software Engineer | Dassault Systems, India | C/C++, Python, Spark, Data Analytics, Linux Apr 2020 - July 2022

- Fixed 10+ critical memory issues in CATIA applications using **C/C++**, enhancing system stability and reducing crash rates by 20%
- Migrated CATIA applications to **Linux** by resolving 20+ errors, cutting migration downtime by 40% and ensuring seamless integration
- Automated CADAM testing using **Python**, reducing test execution time by over 87%, from 15 minutes to 2 minutes workflow

Project Assistant | National Aerospace Laboratories, India | Excel, Tableau, CATIA V5, Python Dec 2019 - Mar 2020

- Pioneered **data analytics** for a VTOL UAV project, boosting efficiency by 17% with **statistical methods** and **data visualization**
- Engineered **interactive visualizations** using **Tableau**, **Matplotlib**, and **Seaborn**, boosting the modeling efficiency by approx. 12.83%
- Optimized 12 propeller blueprints using CATIA V5 and 3D scans, elevating both the **quality** and **performance** of the propeller design

Programmer Analyst | Cognizant, India | Python, Pandas, Machine Learning Dec 2018 - Oct 2019

- Collaborated across teams to implement a **Linear Regression** base model for predicting loan repayments, improving accuracy by 27%
- Consulted with **financial clients** to design **predictive models**, leveraging credit history data to improve loan recovery rates by 12.81%
- Worked closely with **stakeholders** to implement **risk algorithms**, delivering insights that resulted in a \$180,000 gain in the first year

EDUCATION

New Jersey Institute of Technology | MS in Data Science Sep 2022 - Dec 2023

Visvesvaraya Technological University | BE in Mechanical Engineering Aug 2014 - Jul 2018

PROJECTS

Scalable Real-time Content Recommendation Engine | AWS, Spark, Airflow, Kafka

- Engineered a recommendation engine with **AWS EMR** and **Spark**, processing 100GB+ data daily and achieving sub-second response times.
- Reduced data latency by 72% and improved freshness by 4 hrs using **Airflow pipelines**, boosting engagement with timely recommendations
- Built an **AWS Lambda API** to handle 50,000+ requests per second with 99.99% uptime, ensuring seamless **scalability under high demand**

Predictive Healthcare Analytics Platform | AWS SageMaker, Spark ML, Tableau

- Developed models **predicting** patient readmission (**AUC-ROC: 0.87**) and medication adherence, improving early interventions by 25%
- Processed 50GB+ patient data using **Spark ML**, reducing model training time by 71.24% compared to traditional methods
- Created **Tableau** dashboards for healthcare providers, leading to a 15% reduction in preventable hospital readmissions

Intelligent Financial Document Analysis System | RAG, AWS, Spark, Elasticsearch

- Developed a **RAG** system to analyze 1M+ financial documents daily, boosting **insight generation** speed by 58% and enhancing efficiency
- Created custom **financial embeddings**, increasing retrieval accuracy by 37%, enabling precise analysis tailored to financial contexts
- **Integrated GPT-4** with domain-specific prompts, improving insight relevance by 40% and saving 10+ hrs of manual analysis weekly

Gravitational Wave Detection | Python, Time Series Analysis, TensorFlow/Keras, PySpark

- Innovated a robust **deep learning model** using **TensorFlow** and **Keras** for Gravitational Wave Detection, achieving an accuracy of 86.84%
- Processed 76GB+ of LIGO/VIRGO **time-series data** with **PySpark**, overcoming memory limits to enable **large-scale analysis**
- Implemented **Neural Architecture** combining EfficientNetB7 for feature extraction with ANNs for precise **binary classification**